Dynamic Event-Triggered and Self-Triggered Fault-Tolerant Attitude Control for Multiple Spacecraft Systems With Uncertainties and Input Saturation

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This article proposes event-triggered fault-tolerant attitude control strategies for nonlinear multispacecraft systems with disturbances, uncertainties, and input constraints. First, a dynamic event-based fault-tolerant attitude control algorithm is proposed to avoid continuous communication, where the dynamic and constant parameters are introduced to enlarge the trigger threshold, which improves resource conservation efficiency. Second, a self-triggered fault-tolerant attitude controller is constructed to avert continuous detection of events, indicating that continuous computing is not required and the computational burden can be reduced. Moreover, the proposed trigger mechanisms make the system have a positive interevent time, which shows the system is Zeno-free. Finally, simulations are provided to demonstrate the effectiveness of the developed controllers.

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I. INTRODUCTION

Recently, the working mode of using multiple small satellites to form a spacecraft formation flying (SFF) system has attracted special interest [1], [2], [3], [4]. In order to realize high system performance and carry out the goal of replacing large satellites, formation spacecraft usually need to maintain a specific formation configuration. Maintaining formation configuration is the foundation for accomplishing formation missions. Therefore, it is necessary to maintain consistency in the state changes of the formation spacecraft by communicating with neighbors to retain the structure, which is the main difference between the formation control algorithm and the single satellite control algorithm [5], [6], [7].

Robustness is a primary index for designing control systems. Note that satellites are inevitably involved by perturbations while in orbit. To work out this problem, some robust formation control strategies are given by scholars [8], [9]. For example, a robust control law was proposed in [10] to make the formation members have the robustness to disturbances and uncertainties, where the dynamic function was introduced to optimize control parameters. In [11], a fast control method was designed to handle the nonlinearities and uncertainties of formation satellites, and the formation control tasks were realized in finite time.

Despite the rapid development of aerospace microelectronics technology, communication and computing resources on spacecraft are still very limited. The above controllers need to be based on continuous communication, which is a challenge for the control system of small spacecraft. To reduce the communication burden, the common method is time-triggered, i.e., sampling. Nevertheless, this method lacks flexibility and scalability, resulting in low efficiency in resource conservation [12]. To improve the efficiency in saving resources, the event-triggered control mechanism is proposed in [13], [14], [15], and [16]. For first-order formation systems, a distributed event-based control policy was developed and a trigger function using itself state and neighbors sampling states was established in [17]. A fixed-time control approach was provided for multiple stratospheric airship systems, and the event-triggered control was introduced to cut down the calculation frequency of control inputs in [18]. The presence of disturbances makes the event-triggered control design complex and difficult. For nonlinear formation systems with disturbances, the predefined-time formation control problem under limited resources was discussed and the event-driven control strategy was adopted to avoid continuous communication in [19]. The finite-time event-triggered formation control problem was investigated in [20], where an observer using the trigger information was established to estimate disturbances, and the triggering condition was designed to diminish the update frequency of the control law. The trigger mechanisms proposed in [17], [18], [19], and [20] utilize state information as the triggering threshold. To obtain a larger trigger threshold and further reduce the update frequency, dynamic parameters are introduced [21], [22]. In [23], a dynamic trigger function was presented to save resources, and a dynamic event-driven formation-containment control approach was provided to accomplish formation purposes under limited resources. A dynamic parameter and an exponential term were introduced into the triggering mechanism to conserve communication bandwidth in [24]. Note that although the above triggering mechanisms reduce the number of controller calculations, they need to introduce a continuous detection device to determine whether the triggering conditions are met, causing additional computational burden. Considering the control input is constant between two events, it is feasible to estimate the next trigger time according to the current information. Therefore, the self-triggered control method is developed to avoid continuous detection, meaning continuous computation does not require [25], [26], [27]. For linear multispacecraft systems, a self-triggered attitude control structure was configured to economize communication and computation resources in [28]. For nonlinear multiple stratospheric airship systems, a self-triggered control strategy was developed in [29] to utilize limited resources to deal with disturbances, saturation, and delays.

Unlike other formation systems, SFF systems operate in harsh space environments, leading to a high probability of actuator failure. In addition, the control inputs that spacecraft can provide are limited. As a consequence, it is necessary to consider actuator failures and saturation in the design of spacecraft formation control algorithms. To address the formation control problem under actuator faults and input saturation, some fault-tolerant control algorithms and antisaturation control algorithms are given [30], [31]. For single spacecraft systems with disturbances, input constraints, and actuator faults, an event-based fault-tolerant attitude control policy was proposed in [32] to realize fast convergence while saving communication resources. In [33], the event-based control problem for linear spacecraft formation systems with saturated inputs was studied, a dynamic event-triggered control method was presented to avoid continuous communication, and a self-triggered control strategy was designed to further avoid continuous computation. An event-driven fault-tolerant control law was established for spacecraft formation systems subject to disturbances in [34], where a trigger condition using triggering states was proposed to lower the communication burden. There are many studies on event-triggered control strategies, but there are almost no self-triggered control algorithms for nonlinear spacecraft formation systems, which is the main motivation of this article.

As mentioned, this article investigates the fault-tolerant attitude control problem for SFF subject to limited resources, input constraints, and disturbances. A dynamic event-based attitude control approach is proposed to conserve communication bandwidth. The introduction of dynamic and constant parameters increases the triggering threshold, which enlarges the triggering interval. Moreover, a self-triggered mechanism is established to save computation resources. The system has good control performance under the self-triggered controller. The contributions of this work are summarized as follows.

1) A dynamic event-triggered fault-tolerant attitude control law is proposed for SFF systems. Compared with the dynamic event-triggered controllers given in [23] and [34], the introduction of the constant parameter increases the triggering threshold and reduces communication frequency. In addition, the consideration of actuator failures and input saturation makes the proposed algorithm more in line with the actual requirements of formation spacecraft.
2) For nonlinear spacecraft formation systems, a self-triggered fault-tolerant control algorithm is designed to avoid continuous detection and decrease computation burden. In contrast to the self-triggered controllers developed in [28] and [29], a novel self-triggered design idea is given for nonlinear formation systems.
3) Unlike traditional control strategies, the developed dynamic event-driven control approach can efficiently conserve communication resources. Compared with the dynamic event-based control method, the proposed self-triggered controller can avoid continuous computing, improve convergence accuracy, but increase the number of triggers.

The rest of the article is organized as follows. Preliminaries are given in Section II. The dynamic event-triggered and self-triggered control algorithms are presented in Section III. In Section IV, simulation results are provided to verify the theoretical results. Finally, Section V concludes this article.

II. PRELIMINARIES

A. Graph Theory

The communication topology among spacecraft is denoted by a directed graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ with the spacecraft set $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. If $(v_i, v_j) \in \mathcal{E}$, v_j can obtain information from v_i, but not vice versa. Denote the adjacency matrix by $\boldsymbol{A} = [a_{ij}] \in \mathbb{R}^{n\times n}$. If $(v_j, v_i) \in \mathcal{E}$, $a_{ij} = 1$, otherwise $a_{ij} = 0$. Besides, $a_{ii} = 0$. The desired adjacency matrix is depicted by $\boldsymbol{B} = \mathrm{diag}\{b_1, b_2, \ldots, b_n\} \in \mathbb{R}^{n\times n}$. If v_i has access to the desired information, $b_i = 1$, otherwise $b_i = 0$.

ASSUMPTION 1 The graph $\mathcal{G}$ contains a spanning tree, and the root spacecraft can receive the desired information.

B. Problem Description

Considering the formation spacecraft with actuator faults and input constraints, the attitude dynamics of the ith spacecraft are described as

$$\dot{\boldsymbol{\sigma}}_i = \boldsymbol{G}(\boldsymbol{\sigma}_i)\,\boldsymbol{\omega}_i \tag{1}$$

$$\boldsymbol{J}_i\dot{\boldsymbol{\omega}}_i = -\boldsymbol{\omega}_i^{\times}\boldsymbol{J}_i\boldsymbol{\omega}_i + \boldsymbol{\Gamma}_i\mathrm{sat}(\boldsymbol{u}_i) + \bar{\boldsymbol{u}}_i + \boldsymbol{\rho}_i \tag{2}$$

in which $\boldsymbol{G}(\boldsymbol{\sigma}_i) = 0.25(1 - \|\boldsymbol{\sigma}_i\|^2)\boldsymbol{I}_3 + 0.5\boldsymbol{\sigma}_i^\times + 0.5\boldsymbol{\sigma}_i\boldsymbol{\sigma}_i^T$, $\boldsymbol{I}_3 \in \mathbb{R}^{3\times 3}$ is the identity matrix, $\|\boldsymbol{\sigma}_i\|$ denotes the two-norm of $\boldsymbol{\sigma}_i$, $\boldsymbol{\sigma}_i \in \mathbb{R}^3$ is the attitude expressed by Modified Rodrigues Parameters, $\boldsymbol{\omega}_i \in \mathbb{R}^3$ is the angular velocity, $\boldsymbol{J}_i \in \mathbb{R}^{3\times 3}$ is the inertia matrix, $\boldsymbol{\Gamma}_i = \mathrm{diag}(\Gamma_{i1}, \Gamma_{i2}, \Gamma_{i3}) \in \mathbb{R}^{3\times 3}$ with $\Gamma_{ij} \in (0, 1]$, $j = 1, 2, 3$ is the effectiveness matrix, $\bar{\boldsymbol{u}}_i \in \mathbb{R}^3$ is the drift torque, $\boldsymbol{u}_i \in \mathbb{R}^3$ is the control input, $\boldsymbol{\rho}_i \in \mathbb{R}^3$ is the lumped disturbance including external disturbances and uncertainties. Furthermore, for $\boldsymbol{x} = [x_1, x_2, x_3]^T \in \mathbb{R}^3$, $\boldsymbol{x}^\times$ denotes

$$\boldsymbol{x}^\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

$\mathrm{sat}(\boldsymbol{u}_i) = [\mathrm{sat}(u_{ix}), \mathrm{sat}(u_{iy}), \mathrm{sat}(u_{iz})]^T \in \mathbb{R}^3$, where $\mathrm{sat}(u_{ij}) = \min(|u_{ij}|, u_{\max})\mathrm{sgn}(u_{ij})$, $j = x, y, z$, $u_{\max}$ and $\mathrm{sgn}(\cdot)$ are the maximum control input and the signum function, respectively. Therefore, we can obtain $\mathrm{sat}(\boldsymbol{u}_i) = \boldsymbol{h}_i\boldsymbol{u}_i$, where $\boldsymbol{h}_i = \mathrm{diag}(h_{i1}, h_{i2}, h_{i3})$ and h_{ij} is given by

$$h_{ij} = \begin{cases} 1, & |u_{ij}| < u_{\max} \\ \frac{u_{\max}\mathrm{sgn}(u_{ij})}{u_{ij}}, & |u_{ij}| \geq u_{\max} \end{cases}, \ j = 1, 2, 3.$$

According to the definition of h_{ij}, it is obvious that $0 < h_{ij} \leq 1$.

The objective of this article is to develop an event-based fault-tolerant attitude formation control policy for SFF systems such that attitude consensus can be achieved in the presence of actuator failures, input constraints, and limited resources.

III. MAIN RESULTS

For $t \in [t_k^i, t_{k+1}^i)$, define three variables to denote the formation system errors

$$\boldsymbol{e}_{1i} = \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\sigma}_i - \boldsymbol{\sigma}_j\left(t_k^j\right)\right) + b_i(\boldsymbol{\sigma}_i - \boldsymbol{\sigma}_d) \quad (3)$$

$$\boldsymbol{e}_{2i} = \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\omega}_i - \boldsymbol{\omega}_j\left(t_k^j\right)\right) + b_i(\boldsymbol{\omega}_i - \boldsymbol{\omega}_d) \quad (4)$$

$$\boldsymbol{s}_i = \boldsymbol{e}_{2i} + r\boldsymbol{e}_{1i} \quad (5)$$

where t_k^i is the latest trigger instant of ith spacecraft, t_{k+1}^i is the next trigger instant, t_k^j is the latest triggering time of jth spacecraft at the instant t_k^i and $t_k^j \leq t_k^i < t_{k+1}^j$, $\boldsymbol{\sigma}_d$ is the expected attitude, $\boldsymbol{\omega}_d$ is the expected angular velocity, and $r > 0$.

Combining (2)–(5), one obtains

$$\begin{aligned}\dot{\boldsymbol{s}}_i &= \dot{\boldsymbol{e}}_{2i} + r\dot{\boldsymbol{e}}_{1i} \\ &= \left(\sum_{j=1}^{n} a_{ij} + b_i\right)\boldsymbol{J}_i^{-1}\left(-\boldsymbol{\omega}_i^\times\boldsymbol{J}_i\boldsymbol{\omega}_i + \boldsymbol{\Gamma}_i\boldsymbol{h}_i\boldsymbol{u}_i + \bar{\boldsymbol{u}}_i + \boldsymbol{\rho}_i\right) \\ &\quad - b_i\dot{\boldsymbol{\omega}}_d + r\dot{\boldsymbol{e}}_{1i}. \end{aligned} \quad (6)$$

Further, define $\boldsymbol{f}_i = (\sum_{j=1}^{n} a_{ij} + b_i)(\bar{\boldsymbol{u}}_i + \boldsymbol{\rho}_i - \boldsymbol{\omega}_i^\times\boldsymbol{J}_i\boldsymbol{\omega}_i) + \boldsymbol{J}_i(r\dot{\boldsymbol{e}}_{1i} - b_i\dot{\boldsymbol{\omega}}_d)$. Equation (6) can be rewritten as

$$\boldsymbol{J}_i\dot{\boldsymbol{s}}_i = \left(\sum_{j=1}^{n} a_{ij} + b_i\right)\boldsymbol{\Gamma}_i\boldsymbol{h}_i\boldsymbol{u}_i + \boldsymbol{f}_i. \quad (7)$$

ASSUMPTION 2 The nonlinearity term $\boldsymbol{f}_i$ satisfies $\|\boldsymbol{f}_i\| \leq \alpha_i\Phi_i$, where $\alpha_i > 0$ is constant and $\Phi_i = 1 + \|\boldsymbol{\omega}_i(t_k^i)\| + \|\boldsymbol{\omega}_i(t_k^i)\|^2$.

REMARK 1 In light of the definition of $\boldsymbol{f}_i$, it is reasonable to assume $(\sum_{j=1}^{n} a_{ij} + b_i)\|\boldsymbol{\omega}_i^\times\boldsymbol{J}_i\boldsymbol{\omega}_i\| \leq c_{1i}\|\boldsymbol{\omega}_i\|^2$, $\|\boldsymbol{J}_i r(\dot{\boldsymbol{e}}_{1i} + b_i\dot{\boldsymbol{\sigma}}_d)\| \leq c_{2i}\|\boldsymbol{\omega}_i\|$, and $\|(\sum_{j=1}^{n} a_{ij} + b_i)(\bar{\boldsymbol{u}}_i + \boldsymbol{\rho}_i) - \boldsymbol{J}_i b_i(r\dot{\boldsymbol{\sigma}}_d + \dot{\boldsymbol{\omega}}_d)\| \leq c_{3i}$, in which $c_{1i} > 0$, $c_{2i} > 0$, and $c_{3i} > 0$. Thereby, one gets

$$\|\boldsymbol{f}_i\| \leq c_{1i}\|\boldsymbol{\omega}_i\|^2 + c_{2i}\|\boldsymbol{\omega}_i\| + c_{3i}.$$

Furthermore, the change in state between two adjacent triggers is limited, showing that

$$\begin{aligned} c_{1i}\|\boldsymbol{\omega}_i\|^2 + c_{2i}\|\boldsymbol{\omega}_i\| + c_{3i} &\leq c_{1i}\left\|\boldsymbol{\omega}_i\left(t_k^i\right)\right\|^2 + c_{4i} \\ &+ c_{2i}\left\|\boldsymbol{\omega}_i\left(t_k^i\right)\right\| + c_{5i} + c_{3i} \end{aligned}$$

in which $c_{4i} > 0$, and $c_{5i} > 0$. It is further obtained that

$$\|\boldsymbol{f}_i\| \leq c_{1i}\left\|\boldsymbol{\omega}_i\left(t_k^i\right)\right\|^2 + c_{2i}\left\|\boldsymbol{\omega}_i\left(t_k^i\right)\right\| + c_{6i} \leq \alpha_i\Phi_i$$

where $c_{6i} = c_{3i} + c_{4i} + c_{5i}$, and $\alpha_i = \max(c_{1i}, c_{2i}, c_{6i})$.

ASSUMPTION 3 The diagonal matrices $\boldsymbol{\Gamma}_i$ and $\boldsymbol{h}_i$ satisfy $\delta_i \leq \|\boldsymbol{\Gamma}_i\boldsymbol{h}_i\| \leq 1$, in which $\delta_i > 0$.

To save communication resources, an event-based fault-tolerant attitude coordination control structure is designed as follows:

$$\boldsymbol{u}_i = -k_{1i}\boldsymbol{s}_i\left(t_k^i\right) - k_{2i}\alpha_i\Phi_i\mathrm{sgn}\left(\boldsymbol{s}_i\left(t_k^i\right)\right) \quad (8)$$

in which $k_{1i} > 0$, $k_{2i} > 0$, and $\boldsymbol{s}_i(t_k^i)$ is given by

$$\begin{aligned} \boldsymbol{s}_i\left(t_k^i\right) &= \boldsymbol{e}_{2i}\left(t_k^i\right) + r\boldsymbol{e}_{1i}\left(t_k^i\right) \\ \boldsymbol{e}_{1i}\left(t_k^i\right) &= \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\sigma}_i\left(t_k^i\right) - \boldsymbol{\sigma}_j\left(t_k^j\right)\right) + b_i\left(\boldsymbol{\sigma}_i\left(t_k^i\right) - \boldsymbol{\sigma}_d\left(t_k^i\right)\right) \\ \boldsymbol{e}_{2i}\left(t_k^i\right) &= \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\omega}_i\left(t_k^i\right) - \boldsymbol{\omega}_j\left(t_k^j\right)\right) \\ &\quad + b_i\left(\boldsymbol{\omega}_i\left(t_k^i\right) - \boldsymbol{\omega}_d\left(t_k^i\right)\right). \end{aligned}$$

Then, define the following error vector:

$$\tilde{\boldsymbol{s}}_i = \boldsymbol{s}_i\left(t_k^i\right) - \boldsymbol{s}_i. \quad (9)$$

Subsequently, the triggering function is constructed as

$$t_{k+1}^i = \inf\left\{t > t_k^i \mid \|\tilde{\boldsymbol{s}}_i\|^2 - \delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 \geq \chi_i + \varepsilon_i\right\} \quad (10)$$

in which $\varepsilon_i > 0$, χ_i is the dynamic variable and is defined by

$$\dot{\chi}_i = -\beta_{1i}\chi_i + \beta_{2i}\left(\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \varepsilon_i - \|\tilde{\boldsymbol{s}}_i\|^2\right). \quad (11)$$

β_{1i} and β_{2i} are positive constants.

REMARK 2 Only when the trigger condition (10) is satisfied, the event-triggered fault-tolerant attitude cooperative control algorithm (8) will be updated and the formation spacecraft will transmit its triggering state information to adjacent spacecraft. When adjacent spacecraft receive triggering state information, their controllers are not updated. In other words, between two consecutive triggering events, the update of the controller only depends on the triggering condition, while the triggering state information of other spacecraft does not affect the current triggering conditions.

REMARK 3 Unlike the traditional dynamic event-triggered control, the parameter ε_i is introduced to decrease the triggering numbers. When the variable $\boldsymbol{s}_i$ is large, the trigger function requires high-frequency triggering to achieve fast stability, while when the error $\boldsymbol{s}_i$ is small, the trigger function requires low-frequency triggering to save resources. The introduction of ε_i is precisely based on this idea. When $\boldsymbol{s}_i$ is large, the term $\delta_i \|\boldsymbol{s}_i(t_k^i)\|^2 + \chi_i$ plays the leading role in the trigger threshold, while when $\boldsymbol{s}_i$ is small, the term ε_i plays the leading role.

THEOREM 1 For the formation system (1) and (2), if Assumptions 1–3 are satisfied, $\chi_i(0) > 0$, $k_{1i} = 2(\gamma_i \beta_{1i} + \beta_{2i})/(\sum_{j=1}^n a_{ij} + b_i)$, and $k_{2i} = (k_0 + 1)/[(\sum_{j=1}^n a_{ij} + b_i)\delta_i]$, where $0 < \gamma_i < 1$ and $k_0 > 0$. Then, the system states will cooperatively approach to the expected states under the event-triggered fault-tolerant attitude synchronization control algorithm given in (8) and the trigger conditions given in (10) and (11).

PROOF In view of (10) and (11), we get

$$\dot{\chi}_i \geq -\beta_{1i}\chi_i - \beta_{2i}\chi_i. \tag{12}$$

Therefore, $\chi_i(t) \geq \chi_i(0) e^{-(\beta_{1i}+\beta_{2i})t} > 0$.
Choose the Lyapunov function

$$V = \frac{1}{2}\sum_{i=1}^n \boldsymbol{s}_i^T \boldsymbol{J}_i \boldsymbol{s}_i + \sum_{i=1}^n \chi_i. \tag{13}$$

Combining (7), (8), and (13), one obtains

$$\begin{aligned}
\dot{V} &= \sum_{i=1}^n \boldsymbol{s}_i^T \boldsymbol{J}_i \dot{\boldsymbol{s}}_i + \sum_{i=1}^n \dot{\chi}_i \\
&= \sum_{i=1}^n \boldsymbol{s}_i^T \left[\left(\sum_{j=1}^n a_{ij} + b_i\right) \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{u}_i + \boldsymbol{f}_i\right] + \sum_{i=1}^n \dot{\chi}_i \\
&= -\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) + \sum_{i=1}^n \boldsymbol{s}_i^T \boldsymbol{f}_i \\
&\quad + \sum_{i=1}^n \dot{\chi}_i - \sum_{i=1}^n k_{2i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \\
&\quad \times \alpha_i \Phi_i \boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \operatorname{sgn}\left(\boldsymbol{s}_i\left(t_k^i\right)\right).
\end{aligned} \tag{14}$$

If the system trajectories begin from a region in which $\boldsymbol{s}_i^T \operatorname{sgn}(\boldsymbol{s}_i(t_k^i)) = \boldsymbol{s}_i^T \operatorname{sgn}(\boldsymbol{s}_i)$, (14) can be expressed as

$$\begin{aligned}
\dot{V} \leq & -\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) + \sum_{i=1}^n \boldsymbol{s}_i^T \boldsymbol{f}_i \\
& -\sum_{i=1}^n k_{2i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \alpha_i \Phi_i \delta_i \|\boldsymbol{s}_i\| + \sum_{i=1}^n \dot{\chi}_i.
\end{aligned} \tag{15}$$

In accordance with the relationship between $\tilde{\boldsymbol{s}}_i$, $\boldsymbol{s}_i(t_k^i)$, and $\boldsymbol{s}_i$, one obtains

$$\begin{aligned}
\boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) &= \frac{1}{2}\left(\boldsymbol{s}_i\left(t_k^i\right) - \tilde{\boldsymbol{s}}_i\right)^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) \\
&\quad + \frac{1}{2}\boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i\left(\tilde{\boldsymbol{s}}_i + \boldsymbol{s}_i\right) \\
&= \frac{1}{2}\boldsymbol{s}_i^T\left(t_k^i\right) \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) + \frac{1}{2}\boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i \\
&\quad - \frac{1}{2}\tilde{\boldsymbol{s}}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \boldsymbol{s}_i\left(t_k^i\right) + \frac{1}{2}\boldsymbol{s}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i \tilde{\boldsymbol{s}}_i \\
&\geq \frac{1}{2}\delta_i\left(\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \|\boldsymbol{s}_i\|^2\right) \\
&\quad - \frac{1}{2}\tilde{\boldsymbol{s}}_i^T \boldsymbol{\Gamma}_i \boldsymbol{h}_i\left(\boldsymbol{s}_i\left(t_k^i\right) - \boldsymbol{s}_i\right) \\
&\geq \frac{1}{2}\delta_i\left(\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \|\boldsymbol{s}_i\|^2\right) - \frac{1}{2}\|\tilde{\boldsymbol{s}}_i\|^2.
\end{aligned} \tag{16}$$

Substituting (11) and (16) into (15) gives

$$\begin{aligned}
\dot{V} \leq & -\frac{1}{2}\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \delta_i\left(\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \|\boldsymbol{s}_i\|^2\right) \\
& + \frac{1}{2}\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \|\tilde{\boldsymbol{s}}_i\|^2 + \sum_{i=1}^n \alpha_i \Phi_i \|\boldsymbol{s}_i\| \\
& - \sum_{i=1}^n k_{2i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \alpha_i \Phi_i \delta_i \|\boldsymbol{s}_i\| - \sum_{i=1}^n \beta_{1i}\chi_i \\
& + \sum_{i=1}^n \beta_{2i}\left(\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \varepsilon_i - \|\tilde{\boldsymbol{s}}_i\|^2\right) \\
\leq & -\frac{1}{2}\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \delta_i \|\boldsymbol{s}_i\|^2 \\
& + \frac{1}{2}\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right) \varepsilon_i \\
& + \frac{1}{2}\sum_{i=1}^n k_{1i}\left(\sum_{j=1}^n a_{ij} + b_i\right)\left(\|\tilde{\boldsymbol{s}}_i\|^2 - \delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 - \varepsilon_i\right) \\
& - \sum_{i=1}^n \left[k_{2i}\left(\sum_{j=1}^n a_{ij} + b_i\right)\delta_i - 1\right] \alpha_i \Phi_i \|\boldsymbol{s}_i\| \\
& - \sum_{i=1}^n \beta_{1i}\chi_i + \sum_{i=1}^n \beta_{2i}\left(\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2 + \varepsilon_i - \|\tilde{\boldsymbol{s}}_i\|^2\right)
\end{aligned}$$

$$
\begin{aligned}
&\le -\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\delta_i\|\boldsymbol{s}_i\|^2\\
&+\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\varepsilon_i\\
&-\sum_{i=1}^{n}\left[k_{2i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\delta_i-1\right]\\
&\times\alpha_i\Phi_i\|\boldsymbol{s}_i\|-\sum_{i=1}^{n}\beta_{1i}\chi_i\\
&+\sum_{i=1}^{n}\left[\frac{1}{2}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)-\beta_{2i}\right]\\
&\times\left(\|\tilde{\boldsymbol{s}}_i\|^2-\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2-\varepsilon_i\right).
\end{aligned}\tag{17}
$$

Next, according to the definition of k_{1i} and k_{2i}, one gets

$$
\begin{aligned}
\dot{V}&\le -\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\delta_i\|\boldsymbol{s}_i\|^2-\sum_{i=1}^{n}\beta_{1i}\chi_i\\
&+\sum_{i=1}^{n}\gamma_i\beta_{1i}\left(\|\tilde{\boldsymbol{s}}_i\|^2-\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2-\varepsilon_i\right)\\
&+\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\varepsilon_i-\sum_{i=1}^{n}k_0\alpha_i\Phi_i\|\boldsymbol{s}_i\|\\
&\le -\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\delta_i\|\boldsymbol{s}_i\|^2-\sum_{i=1}^{n}k_0\alpha_i\Phi_i\|\boldsymbol{s}_i\|\\
&-\sum_{i=1}^{n}\beta_{1i}\left(1-\gamma_i\right)\chi_i+\frac{1}{2}\sum_{i=1}^{n}k_{1i}\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\varepsilon_i\\
&\le -\frac{1}{2}\sum_{i=1}^{n}k_{1i}\delta_i\|\boldsymbol{s}_i\|^2-\sum_{i=1}^{n}k_0\alpha_i\Phi_i\|\boldsymbol{s}_i\|\\
&-\sum_{i=1}^{n}\beta_{1i}\left(1-\gamma_i\right)\chi_i+\frac{1}{2}(n+1)\sum_{i=1}^{n}k_{1i}\varepsilon_i\\
&\le -\mu V+\eta
\end{aligned}\tag{18}
$$

where

$$
\begin{aligned}
\mu&=\min\left(k_{1\min}\delta_{\min}/J_{\max},\ \beta_{1\min}\left(1-\gamma_{\max}\right)\right)\\
\eta&=0.5\,(n+1)\,k_{1\max}\varepsilon_{\max}\\
k_{1\min}&=\min\left(k_{11},k_{12},\ldots,k_{1n}\right)\\
\delta_{\min}&=\min\left(\delta_1,\delta_2,\ldots,\delta_n\right)\\
J_{\max}&=\max\left(\|\boldsymbol{J}_1\|,\|\boldsymbol{J}_2\|,\ldots,\|\boldsymbol{J}_n\|\right)\\
\beta_{1\min}&=\min\left(\beta_{11},\beta_{12},\ldots,\beta_{1n}\right)\\
\gamma_{\max}&=\max\left(\gamma_1,\gamma_2,\ldots,\gamma_n\right)\\
k_{1\max}&=\max\left(k_{11},k_{12},\ldots,k_{1n}\right)\\
\varepsilon_{\max}&=\max\left(\varepsilon_1,\varepsilon_2,\ldots,\varepsilon_n\right).
\end{aligned}
$$

From (18), one obtains the state errors will cooperatively converge to a region near the origin in the case of $\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i(t_k^i))=\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i)$.

Remark 4 Note that the event-based fault-tolerant control strategy (8) needs to use parameters α_i and δ_i, which are difficult to be estimated accurately. Considering that the essence of the event-triggered control is to appropriately reduce the convergence accuracy to decrease resource consumption, these parameters can be preliminarily estimated and the influence of estimation errors on the system also reduces the convergence accuracy, indicating that the event-driven fault-tolerant control law (8) is easy to implement.

In light of (9) and (10), we have

$$
\|\tilde{\boldsymbol{s}}_i\|^2=\left\|\boldsymbol{s}_i\left(t_k^i\right)-\boldsymbol{s}_i\right\|^2\le\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2+\chi_i+\varepsilon_i.\tag{19}
$$

It is further obtained that

$$
\begin{aligned}
\|\boldsymbol{s}_i\|&\le\left\|\boldsymbol{s}_i-\boldsymbol{s}_i\left(t_k^i\right)\right\|+\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|\\
&\le\left(\delta_i\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|^2+\chi_i+\varepsilon_i\right)^{1/2}+\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|.
\end{aligned}\tag{20}
$$

When the variable $\boldsymbol{s}_i$ converges to a neighborhood near the origin, the prerequisite $\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i(t_k^i))=\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i)$ may not hold true. In this case, the variable $\boldsymbol{s}_i$ will converge to the region given by (20) due to the triggering mechanism (10). Moreover, if the variable $\boldsymbol{s}_i$ remains in a region where $\|\boldsymbol{s}_i\|>(\delta_i\|\boldsymbol{s}_i(t_k^i)\|^2+\chi_i+\varepsilon_i)^{1/2}+\|\boldsymbol{s}_i(t_k^i)\|$, the sign of the variable $\boldsymbol{s}_i$ will keep unchanged, implying that the equation $\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i(t_k^i))=\boldsymbol{s}_i^T\operatorname{sgn}(\boldsymbol{s}_i)$ holds true under the condition that $\|\boldsymbol{s}_i\|>(\delta_i\|\boldsymbol{s}_i(t_k^i)\|^2+\chi_i+\varepsilon_i)^{1/2}+\|\boldsymbol{s}_i(t_k^i)\|$.

According to the above analysis, the system states will cooperatively converge to the reference states under the event-triggered fault-tolerant control structure given in (8) and the trigger conditions given in (10) and (11). □

Next, we prove that there is no Zeno behavior by proving the interevent time $T_k^i=t_{k+1}^i-t_k^i>\tau_i>0$.

Theorem 2 Considering the formation system (1) and (2) under Assumptions 1–3, if the event-based fault-tolerant control policy is designed as (8) and the trigger function is designed as (10) and (11), the Zeno phenomenon will be ruled out and the interevent time T_k^i will have a positive lower bound τ_i.

Proof Taking the derivative of $\|\tilde{\boldsymbol{s}}_i\|$ gives

$$
\frac{\mathrm{d}}{\mathrm{d}t}\|\tilde{\boldsymbol{s}}_i\|\le\left\|\dot{\tilde{\boldsymbol{s}}}_i\right\|=\left\|\dot{\boldsymbol{s}}_i\left(t_k^i\right)-\dot{\boldsymbol{s}}_i\right\|=\|\dot{\boldsymbol{s}}_i\|.\tag{21}
$$

Substituting (7) and (8) into (21), we have

$$
\begin{aligned}
\|\dot{\boldsymbol{s}}_i\|&=\left\|\boldsymbol{J}_i^{-1}\left[\left(\sum_{j=1}^{n}a_{ij}+b_i\right)\boldsymbol{\Gamma}_i\boldsymbol{h}_i\boldsymbol{u}_i+\boldsymbol{f}_i\right]\right\|\\
&\le\left\|\boldsymbol{J}_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|\boldsymbol{s}_i\left(t_k^i\right)\right\|+\left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}.
\end{aligned}\tag{22}
$$

In addition, considering the trigger condition (10), we obtain

$$\|\tilde{s}_i\| \le \left(\delta_i \left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i + \varepsilon_i\right)^{1/2}. \tag{23}$$

Combining (22) and (23), we get

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\|\tilde{s}_i\|^2 &= 2\,\|\tilde{s}_i\|\,\frac{\mathrm{d}}{\mathrm{d}t}\,\|\tilde{s}_i\| \\ &\le 2\left\|J_i^{-1}\right\|\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i + \varepsilon_i\right)^{1/2} \\ &\quad \times \left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}.\end{aligned} \tag{24}$$

For $t \in [t_k^i, t_{k+1}^i)$, $\|\tilde{s}_i\|^2$ grows from zero to $\delta_i\|s_i(t_k^i)\|^2 + \chi_i + \varepsilon_i$. Therefore, the following inequality can be obtained:

$$\begin{aligned}T_k^i &\ge \tau_i \\ &= \frac{\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i + \varepsilon_i\right)^{1/2}}{2\left\|J_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}}.\end{aligned} \tag{25}$$

From Theorem 1, $\tilde{s}_i$ and s_i will approach to a small neighborhood under the event-triggered control approach (8), which indicates $s_i(t_k^i)$ and $\omega_i(t_k^i)$ are bounded. Thus, the interevent time T_k^i has a positive lower bound τ_i, showing that the system is Zeno-free. □

To avoid continuous communication, the event-triggered control method is applied. However, the trigger mechanism requiring continuous detection is introduced, which makes it difficult for the event-based controller to implement and aggravates the computational burden of the system. The next trigger time can be estimated by the information of the current trigger time due to the constant control input between two successive events. Using this idea, a self-triggered attitude coordinated control approach is developed via the event-triggered control protocol (8) to avoid continuous computing.

THEOREM 3 Consider the formation system (1) and (2) under Assumptions 1–3. If the event-triggered control structure is configured as (8), and the trigger time is described in

$$\begin{aligned}t_{k+1}^i &= t_k^i \\ &+ \frac{\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i(0)\,e^{-(\beta_{1i}+\beta_{2i})t_{k+1}^i} + \varepsilon_i\right)^{1/2}}{\left\|J_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}}.\end{aligned} \tag{26}$$

Then, the attitude consensus will be achieved with low resource consumption.

PROOF From (23), $\|\tilde{s}_i\|$ grows from zero to $(\delta_i\|s_i(t_k^i)\|^2 + \chi_i + \varepsilon_i)^{1/2}$ for $t \in [t_k^i, t_{k+1}^i)$. At the next trigger time t_{k+1}^i, $\|\tilde{s}_i\|$ is reset to zero.

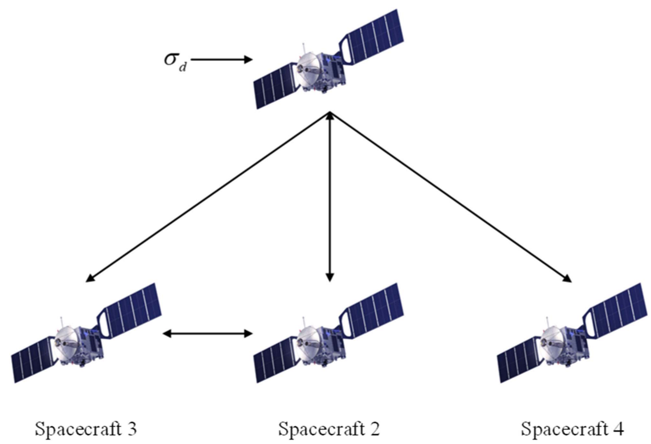

Fig. 1. Communication topology.

Therefore, combining (22), we can obtain

$$\begin{aligned}t - t_k^i &\ge \\ &\frac{\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i(t) + \varepsilon_i\right)^{1/2}}{\left\|J_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}}.\end{aligned} \tag{27}$$

Consider $\chi_i(t) \ge \chi_i(0)e^{-(\beta_{1i}+\beta_{2i})t} > 0$. $e^{-(\beta_{1i}+\beta_{2i})t}$ is a monotone decreasing function. Therefore, we have

$$\chi_i(t) > \chi_i(0)\,e^{-(\beta_{1i}+\beta_{2i})t_{k+1}^i}. \tag{28}$$

Invoking (27) and (28), we can get

$$\begin{aligned}t - t_k^i &> \\ &\frac{\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i(0)\,e^{-(\beta_{1i}+\beta_{2i})t_{k+1}^i} + \varepsilon_i\right)^{1/2}}{\left\|J_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}}.\end{aligned} \tag{29}$$

Note that the right side of (29) is constant for $t \in [t_k^i, t_{k+1}^i)$. Thus, we have

$$\begin{aligned}t_{k+1}^i - t_k^i &> \\ &\frac{\left(\delta_i\left\|s_i\left(t_k^i\right)\right\|^2 + \chi_i(0)\,e^{-(\beta_{1i}+\beta_{2i})t_{k+1}^i} + \varepsilon_i\right)^{1/2}}{\left\|J_i^{-1}\right\|\left\{(n+1)\,k_{1i}\left\|s_i\left(t_k^i\right)\right\| + \left[(n+1)\,k_{2i}+1\right]\alpha_i\Phi_i\right\}}.\end{aligned} \tag{30}$$

From (30), we can conclude the proof is completed. □

IV. SIMULATION RESULTS

Numerical simulations are carried out to show the superiority of the event-triggered controller in saving resources. The communication topology is shown in Fig. 1 and the corresponding matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are

$$\boldsymbol{A} = \begin{bmatrix}0 & 1 & 0 & 0\\ 1 & 0 & 1 & 0\\ 1 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\end{bmatrix}, \boldsymbol{B} = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}.$$

The reference attitude is $\boldsymbol{\sigma}_d = [3 + 2\sin(0.1t),\ 5 + 3\cos(0.2t),\ -2 + \cos(0.1t)]^T \times 10^{-3}$, and the maximum control input is chosen as $u_{\max} = 0.2\text{N} \cdot \text{m}$.

The inertia matrices are chosen as

$$\boldsymbol{J}_1 = \begin{bmatrix} 5.1 & 0 & 0 \\ 0 & 4.8 & 0 \\ 0 & 0 & 5 \end{bmatrix} \text{kg} \cdot \text{m}^2$$
$$\boldsymbol{J}_2 = \begin{bmatrix} 4.9 & 0 & 0 \\ 0 & 4.9 & 0 \\ 0 & 0 & 5.1 \end{bmatrix} \text{kg} \cdot \text{m}^2$$
$$\boldsymbol{J}_3 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5.2 & 0 \\ 0 & 0 & 5.1 \end{bmatrix} \text{kg} \cdot \text{m}^2$$
$$\boldsymbol{J}_4 = \begin{bmatrix} 4.8 & 0 & 0 \\ 0 & 5.1 & 0 \\ 0 & 0 & 4.9 \end{bmatrix} \text{kg} \cdot \text{m}^2.$$

The lumped disturbances are defined as

$$\begin{aligned} \boldsymbol{\rho}_1 &= [2 + \sin(0.1t), 1 + 2\cos(0.2t), \\ &\quad 2 + \sin(0.1t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \boldsymbol{\rho}_2 &= [1 + 2\cos(0.2t), 2 + \sin(0.1t), \\ &\quad 1 + 2\cos(0.2t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \boldsymbol{\rho}_3 &= [3 + 3\cos(0.1t), 2 + \cos(0.1t), \\ &\quad 2 + \sin(0.2t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \boldsymbol{\rho}_4 &= [2 + \sin(0.2t), 1 + 2\sin(0.2t), \\ &\quad 3 + 3\cos(0.1t)]^T \times 10^{-3}\text{N} \cdot \text{m}. \end{aligned}$$

The effectiveness matrices are chosen as

$$\boldsymbol{\Gamma}_1 = \begin{bmatrix} 0.8 & 0 & 0 \\ 0 & 0.7 & 0 \\ 0 & 0 & 0.8 \end{bmatrix}, \boldsymbol{\Gamma}_2 = \begin{bmatrix} 0.6 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 0.7 \end{bmatrix}$$
$$\boldsymbol{\Gamma}_3 = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 0.6 \end{bmatrix}, \boldsymbol{\Gamma}_4 = \begin{bmatrix} 0.7 & 0 & 0 \\ 0 & 0.6 & 0 \\ 0 & 0 & 0.7 \end{bmatrix}.$$

The drift torques are defined as

$$\begin{aligned} \bar{\boldsymbol{u}}_1 &= [0.2 + 0.3\cos(0.2t), -0.3 + 0.1\cos(0.1t), \\ &\quad 0.1 - 0.2\sin(0.2t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \bar{\boldsymbol{u}}_2 &= [-0.3 + 0.1\sin(0.1t), -0.2 + 0.3\cos(0.2t), \\ &\quad 0.3 - 0.3\sin(0.1t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \bar{\boldsymbol{u}}_3 &= [-0.2 + 0.2\sin(0.1t), -0.2 + 0.2\cos(0.3t), \\ &\quad 0.2 - 0.1\cos(0.3t)]^T \times 10^{-3}\text{N} \cdot \text{m} \\ \bar{\boldsymbol{u}}_4 &= [0.1 + 0.3\cos(0.2t), -0.3 + 0.1\sin(0.2t), \\ &\quad 0.1 - 0.2\sin(0.2t)]^T \times 10^{-3}\text{N} \cdot \text{m}. \end{aligned}$$

The initial attitude is chosen as

$$\begin{aligned} \boldsymbol{\sigma}_1(0) &= [0.15, 0.22, -0.15]^T \\ \boldsymbol{\sigma}_2(0) &= [0.08, 0.18, -0.13]^T \\ \boldsymbol{\sigma}_3(0) &= [-0.09, 0.12, 0.18]^T \\ \boldsymbol{\sigma}_4(0) &= [0.12, -0.13, 0.15]^T. \end{aligned}$$

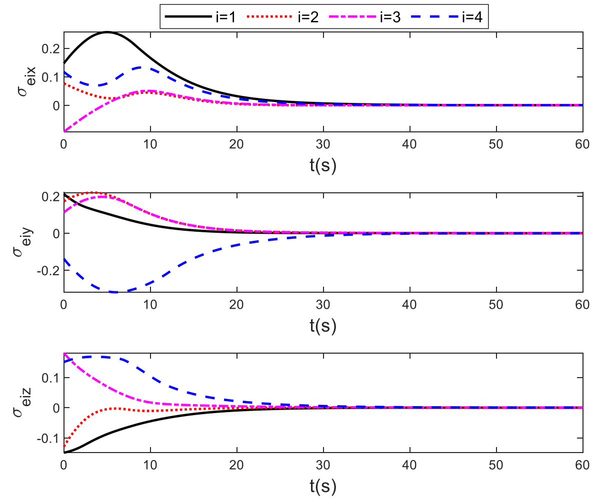

Fig. 2. Attitude errors in Case 1.

The initial angular velocities are given as

$$\begin{aligned} \boldsymbol{\omega}_1(0) &= [0.15, -0.08, 0.12]^T \text{rad/s} \\ \boldsymbol{\omega}_2(0) &= [-0.14, 0.11, 0.13]^T \text{rad/s} \\ \boldsymbol{\omega}_3(0) &= [0.16, 0.12, -0.07]^T \text{rad/s} \\ \boldsymbol{\omega}_4(0) &= [-0.13, -0.17, 0.12]^T \text{rad/s}. \end{aligned}$$

The control parameters are $r = 0.6$, $k_0 = 0.1$, $\alpha_i = 0.001$, $\delta_i = 0.3$, $\varepsilon_i = 0.0001$, $\chi_i(0) = 0.3$, $\beta_{1i} = 5$, $\beta_{2i} = 10$, and $\gamma_i = 0.5$ for $i = 1, 2, 3, 4$. In addition, in light of the definition of k_{1i} and k_{2i}, we have $k_{11} = 12.5$, $k_{12} = 12.5$, $k_{13} = 12.5$, $k_{14} = 25$, $k_{21} = 11/6$, $k_{22} = 11/6$, $k_{23} = 11/6$, and $k_{24} = 11/3$.

The time-driven controller corresponding to the event-driven controller (8) can be defined by

$$\boldsymbol{u}_i = -k_{1i}\boldsymbol{s}_{ti} - k_{2i}\alpha_i \left(1 + \|\boldsymbol{\omega}_i\| + \|\boldsymbol{\omega}_i\|^2\right) \text{sgn}(\mathbf{s}_{ti}) \quad (31)$$

where

$$\begin{aligned} \boldsymbol{s}_{ti} &= \boldsymbol{e}_{2ti} + r\boldsymbol{e}_{1ti} \\ \boldsymbol{e}_{1ti} &= \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\sigma}_i - \boldsymbol{\sigma}_j\right) + b_i\left(\boldsymbol{\sigma}_i - \boldsymbol{\sigma}_d\right) \\ \boldsymbol{e}_{2ti} &= \sum_{j=1}^{n} a_{ij}\left(\boldsymbol{\omega}_i - \boldsymbol{\omega}_j\right) + b_i\left(\boldsymbol{\omega}_i - \boldsymbol{\omega}_d\right). \end{aligned}$$

The following three simulation cases are provided to show the superiority of the proposed event-triggered control strategies.

Case 1: The time-driven controller (31) with the above simulation conditions.

Case 2: The event-driven controller (8) and the trigger mechanism (10) and (11) with the above simulation conditions.

Case 3: The self-triggered controller (8) and the trigger mechanism (26) with the above simulation conditions.

TABLE I
Convergence Accuracy

	Case 1		Case 2		Case 3	
	Attitude errors	Angular velocity errors	Attitude errors	Angular velocity errors	Attitude errors	Angular velocity errors
The X-axis convergence accuracy	1.7×10^{-3}	4.1×10^{-4}	2.6×10^{-3}	9.6×10^{-3}	2.1×10^{-3}	2.1×10^{-3}
The Y-axis convergence accuracy	1.4×10^{-3}	7.1×10^{-4}	5.4×10^{-3}	1.2×10^{-2}	3.6×10^{-3}	3.8×10^{-3}
The Z-axis convergence accuracy	7.7×10^{-4}	2.5×10^{-4}	5.1×10^{-3}	9.7×10^{-3}	1.6×10^{-3}	2.5×10^{-3}

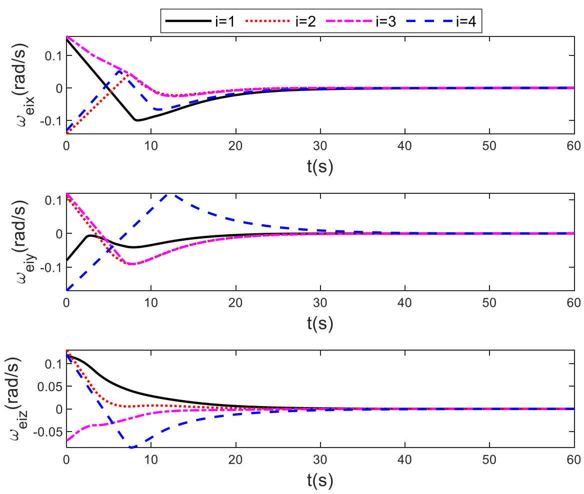

Fig. 3. Angular velocity errors in Case 1.

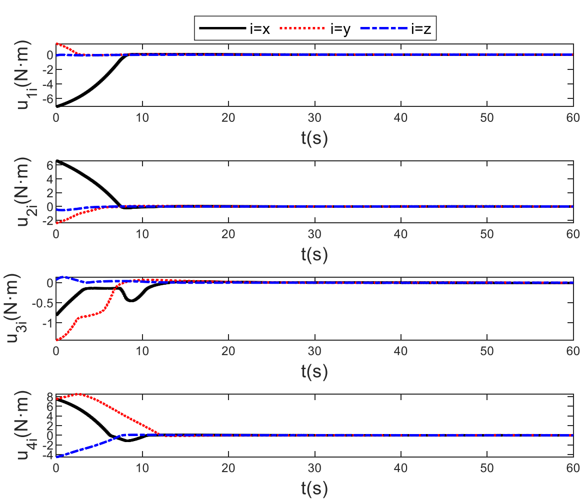

Fig. 4. Control inputs in Case 1.

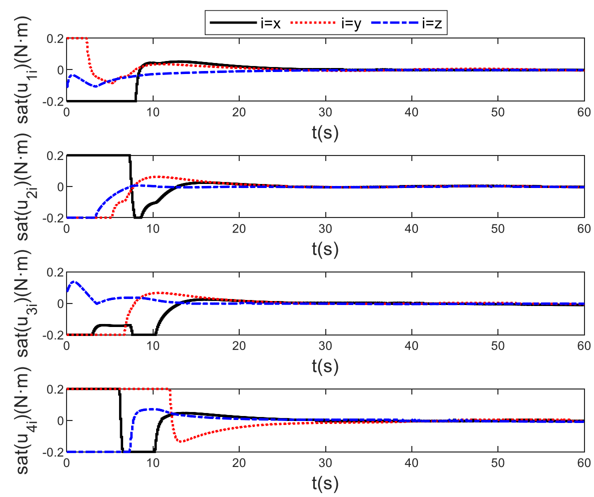

Fig. 5. Control inputs with amplitude constraints in Case 1.

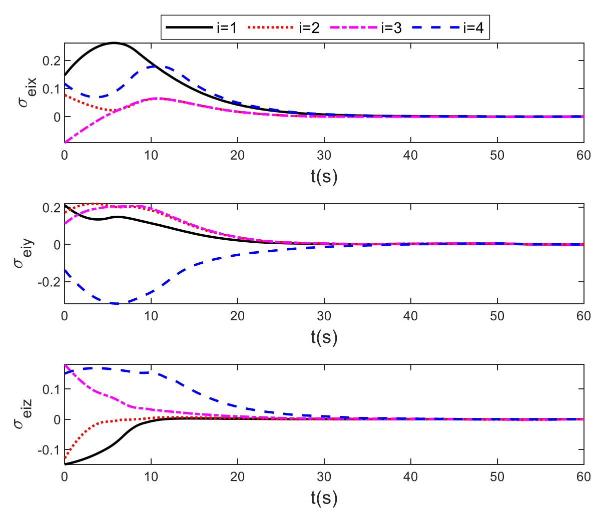

Fig. 6. Attitude errors in Case 2.

Let $\boldsymbol{\sigma}_{ei} = \boldsymbol{\sigma}_i - \boldsymbol{\sigma}_d$ be the attitude error and $\boldsymbol{\omega}_{ei} = \boldsymbol{\omega}_i - \boldsymbol{\omega}_d$ be the angular velocity error. Simulation results are given in Figs. 2–15. Figs. 2, 6, and 11 depict the attitude errors of four formation spacecraft. Figs. 3, 7, and 12 show the angular velocity errors. Figs. 4, 8, and 13 illustrate the control inputs, and Figs. 5, 9, and 14 present the control inputs with amplitude constraints. Figs. 10 and 15 record triggering instants. In addition, the convergence accuracy and trigger times are shown in Tables I and II, respectively. According to the simulation results, the convergence rates of state errors under the three control strategies are almost the same, and state errors can cooperatively converge to the neighborhood

TABLE II
Trigger Times (0–60 s)

	Spacecraft 1	Spacecraft 2	Spacecraft 3	Spacecraft 4
Trigger times in Case 1	600	600	600	600
Trigger times in Case 2	113	116	111	92
Trigger times in Case 3	201	205	203	271

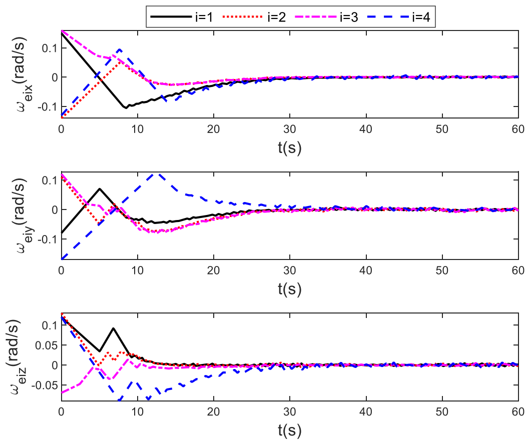

Fig. 7. Angular velocity errors in Case 2.

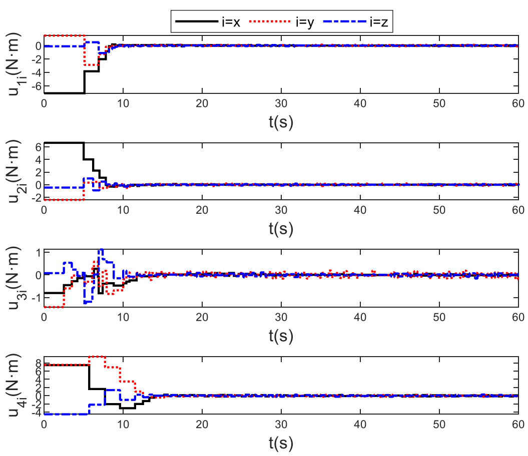

Fig. 8. Control inputs in Case 2.

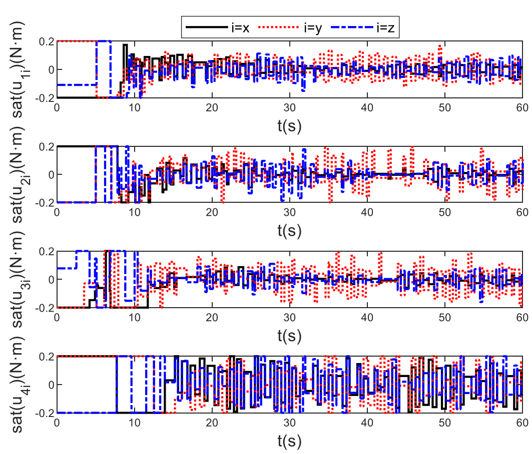

Fig. 9. Control inputs with amplitude constraints in Case 2.

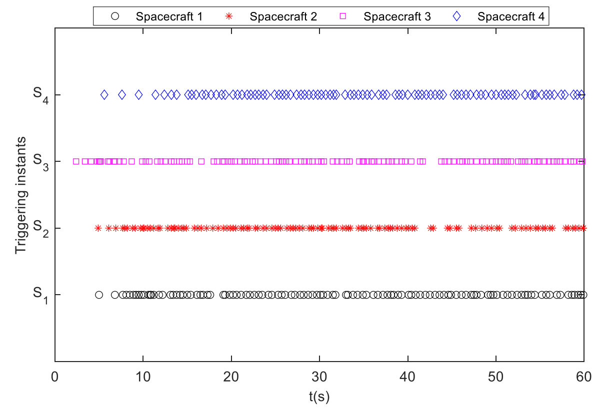

Fig. 10. Triggering instants in Case 2.

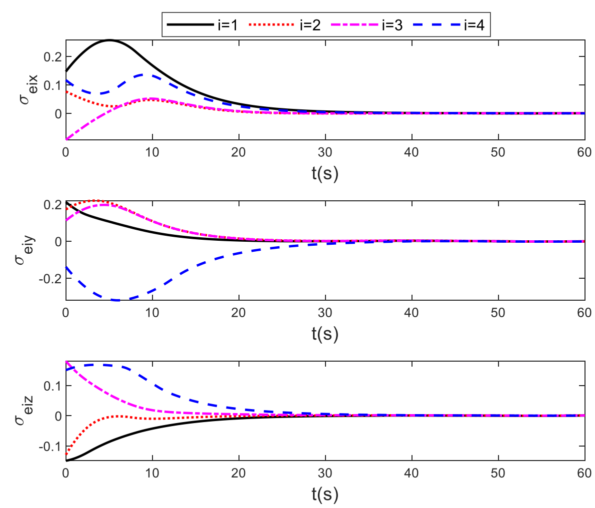

Fig. 11. Attitude errors in Case 3.

around the origin within 40 s. The performance comparison of the three controllers is listed in Table III. For Case 1, the time-driven control algorithm operates at a frequency of 10 Hz, showing the time-driven control law is triggered 600 times in 0–60 s. For Case 2, the trigger frequency of the event-driven control policy is decreased by over 80% by introducing a dynamic triggering function. However, the introduction of the dynamic triggering function reduces the convergence accuracy by almost one order of magnitude. The trigger mechanism also needs to be continuously monitored to determine whether the controller needs to be updated. For Case 3, compared to Case 2, the self-triggered control method is adopted to avoid continuous computation. Although the update numbers are increased by over 100%, the convergence error is decreased by nearly one order of

TABLE III
Performance Comparison of Three Controllers

	Convergence accuracy	Communication resources conservation	Computation resources conservation
The time-driven controller (31)	high	low	low
The event-driven controller (8) with the trigger mechanism (10) and (11)	low	high	low
The self-triggered controller (8) with the trigger mechanism (26)	medium	medium	high

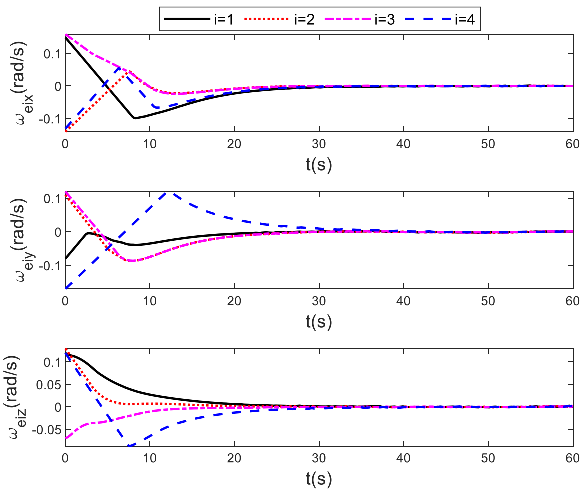

Fig. 12. Angular velocity errors in Case 3.

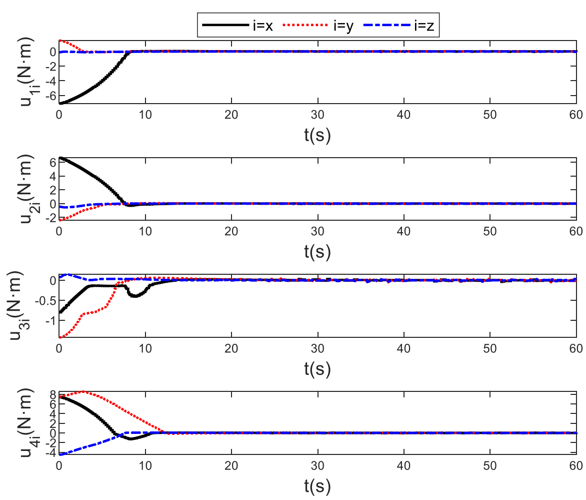

Fig. 13. Control inputs in Case 3.

magnitude. Compared to Case 1, the trigger times can be reduced by 60% and the convergence accuracy is decreased slightly. In addition, the self-triggered control law does not require continuous communication and computation compared to the time-driven control protocol.

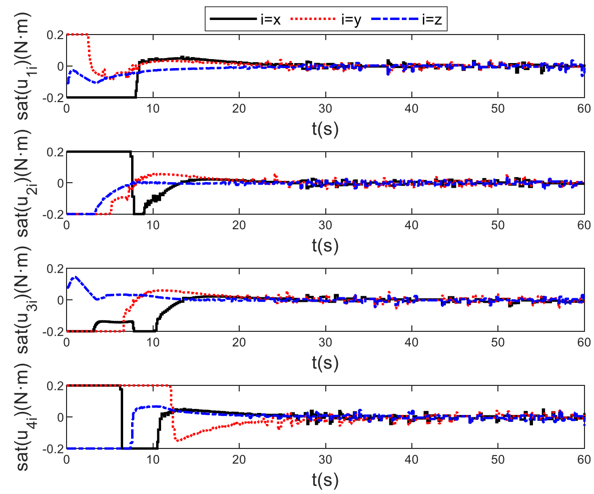

Fig. 14. Control inputs with amplitude constraints in Case 3.

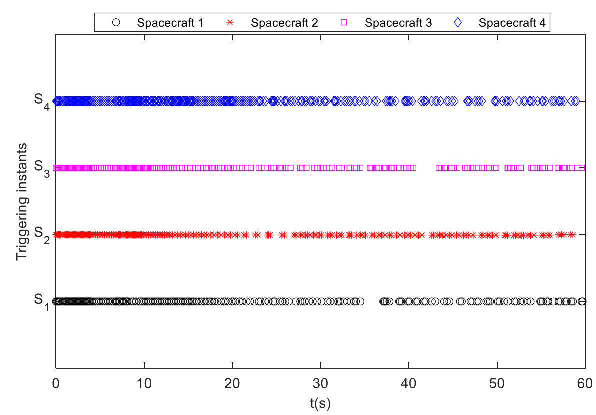

Fig. 15. Triggering instants in Case 3.

V. CONCLUSION

In this article, the fault-tolerant attitude control problem is addressed for nonlinear spacecraft formation systems subject to limited resources, input saturation, disturbances, and uncertainties. To save communication resources, a dynamic event-triggered fault-tolerant control approach is developed and the dynamic and constant parameters are introduced to further decrease the communication burden. Unlike the corresponding continuous controller, the developed controller can significantly reduce the communication frequency with neighbors. To save computation resources, a self-triggered fault-tolerant attitude control mechanism is established. Compared with the dynamic event-driven controller, the developed self-triggered controller can avoid continuous computation, have smaller convergence errors, and have a higher trigger frequency. Furthermore, the Zeno phenomenon is avoided under the proposed event-triggered control laws. Finally, simulations indicate the effectiveness of the developed controllers. Future work will take communication delays into consideration.

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